

ALTEGRAD - MVA 2026

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1 Introduction

1.1 Dataset Overview

The dataset consists of approximately **33,000 molecule-text pairs**. It is partitioned into:

- A **Training and Validation set** of 32,000 samples, where both the graph structure and the ground-truth textual description are provided.
- A **Test set** of 1,000 molecules provided solely as graphs, used for the leaderboard evaluation.

The molecular data is structured as graphs where nodes represent atoms and edges represent chemical bonds. These graphs are enriched with specific attributes: 9 categorical features for atoms (e.g., atomic number, chirality, hybridization) and 3 for bonds (bond type, stereochemistry, conjugation), requiring a specialized encoding strategy.

1.2 Evaluation Metrics

Since this is a generative task, performance is evaluated using a composite score of two complementary Natural Language Processing (NLP) metrics:

- **BLEU-4 Score:** Measures the precision of generated text by calculating the overlap of n-grams (up to 4-grams) between the candidate caption and the reference. It primarily assesses lexical and syntactic accuracy.
- **BERTScore (RoBERTa):** Evaluates the semantic similarity between the generated text and the ground truth using contextual embeddings. This metric ensures that the generated description conveys the correct chemical meaning, even if the phrasing differs from the reference.

2 Explored directions

2.1 "From-scratch" transformer architecture

In parallel to Large Language Model (LLM) based approaches, we implemented a lighter, end-to-end architecture combining a Graph Neural Network encoder with a standard Transformer decoder [3]. Our motivation was twofold: to test if a simpler model could capture the repetitive syntax of molecular captions, and to optimize BLEU scores by restricting the vocabulary via a custom BPE tokenizer, thus avoiding penalizing synonyms. Although the model showed rapid convergence, performance quickly

plateaued. The generated text suffered from elementary grammatical errors (e.g., article confusion "a"/"an"), demonstrating that the broad linguistic knowledge embedded in pre-trained LLMs is essential to maintain fluency, even within a constrained domain.

2.2 CLIP for RAG & LLM

We also explored a Retrieval-Augmented Generation (RAG) strategy to ground the generation process. Using a contrastive pre-training objective similar to CLIP [4], we aligned graph and text embeddings to create a retrieval database. During inference, the caption of the nearest neighbor in the embedding space was retrieved and provided to GPT-2 as a prompt, alongside the target molecule’s representation. The goal was to limit hallucinations by providing a relevant template. However, the model exhibited an over-reliance on the retrieved context: it effectively ignored the target graph embedding and merely regurgitated the retrieved caption. This lack of adaptation to the specific features of the new molecule resulted in low performance scores.

2.3 Editing

We also explored an explicit caption editing approach aimed at correcting chemically invalid tokens in generated/retrieved descriptions. This strategy relied on a custom tokenizer and a restricted vocabulary to detect out-of-vocabulary tokens, which primarily corresponded to rare chemical formulas or molecular identifiers. Detected tokens were then targeted by a dedicated correction model designed to replace incorrect formulas with graph-consistent alternatives. In practice, this approach proved difficult to stabilize: molecular captions follow rigid syntactic templates, and local edits often disrupted grammatical structure. Moreover, some erroneous formulas were structurally far from the true molecular formula, requiring extensive rewriting rather than local correction. As a result, this editing-based approach was ultimately abandoned in favor of methods that better preserve global caption structure.

3 Proposed Method I : End-to-End Graph-to-Sequence Generation

To address the Molecular Graph Captioning task, we developed a multimodal architecture bridging the gap between graph representation learning and pre-trained language models. Our final approach treats the problem as a translation task where the source language is the molecular graph and the target language is the chemical descrip-

tion.

3.1 Rich Molecular Graph Representation

Standard Graph Neural Networks often rely on minimal atomic features. However, accurate chemical captioning requires capturing subtle electronic and geometric properties. We implemented a Multi-Feature Embedding strategy where each atom and bond is described by a combination of categorical properties.

3.1.1 Atom and Bond Embeddings

For each atom v , we do not use a single input vector. Instead, we utilize distinct learnable embedding layers for 9 specific chemical attributes: *Atomic Number*, *Chirality Tag*, *Degree*, *Formal Charge*, *Number of Hydrogens*, *Radical Electrons*, *Hybridization*, *Aromaticity*, and *Ring Membership*. Mathematically, the initial feature vector $h_v^{(0)}$ is obtained by the element-wise summation of these specific embeddings:

$$h_v^{(0)} = \sum_{k \in \mathcal{F}_{atom}} \text{Emb}_k(x_{v,k}) \quad (1)$$

where $\mathcal{F}_{atom}$ is the set of atomic features. This summation strategy allows the model to learn complex interactions without exploding the dimensionality.

Similarly, edges (bonds) are encoded via a **MultiFeatureBondEncoder** that aggregates *Bond Type* (single, double, triple, aromatic), *Stereo-configuration* (E/Z), and *Conjugation status*. These edge features e_{uv} are critical for distinguishing stereoisomers, which directly impacts the IUPAC name generation.

3.2 Sequence Length Analysis and Tokenization

Transformer-based models have a memory complexity quadratic to the sequence length $O(L^2)$. Selecting an optimal maximum sequence length (L_{max}) is therefore a trade-off between computational efficiency and information coverage.

We conducted a statistical analysis of the tokenized captions in the training and validation sets using the T5 tokenizer (see Table 3 in Appendix). The analysis revealed a high variance in caption length, ranging from 29 to 920 tokens. However, the distribution is heavily skewed:

- The median length is approximately 100 tokens.
- The 99th percentile of the validation set is 274.14 tokens.

Based on these observations, we fixed $L_{max} = 274$. This threshold covers 99% of the validation samples without truncation, while significantly reducing memory usage compared to a "safe" maximum of 512 or 1024 tokens.

3.3 Structural Encoder: Deep GINE

We selected the **Graph Isomorphism Network with Edge Features (GINE)** [1] as our encoder backbone. Unlike standard GCNs, GINE explicitly incorporates edge

attributes into the message-passing aggregation, which is essential for chemistry. Our encoder consists of a stack of 10 GINE layers. At each layer l , the update rule is:

$$h_v^{(l)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)}) \cdot h_v^{(l-1)} + \sum_{u \in \mathcal{N}(v)} \text{ReLU}(h_u^{(l-1)} + e_{uv}) \right) \quad (2)$$

To facilitate gradient flow across this deep architecture, we employed Residual Connections ($h^{(l)} = h_{GINE}^{(l)} + h^{(l-1)}$) and Layer Normalization after each block. This design prevents the over-smoothing problem common in deep GNNs and ensures that structural information is preserved up to the bottleneck layer.

3.4 Decoder and Training Strategy

We interface the graph encoder with a pre-trained **T5-Small** decoder, drawing inspiration from MolT5 [2]. Since T5 expects a sequence of continuous embeddings as input (normally coming from its own text encoder), we employ a Soft-Prompting approach to bridge the modality gap between the graph latent space and the language model.

3.4.1 Modality Alignment via Linear Projection

The output of the Graph Encoder is a set of node embeddings $H_G \in \mathbb{R}^{N \times d_{graph}}$, where N is the number of nodes and $d_{graph} = 512$. To align these representations with the T5 hidden dimension ($d_{T5} = 512$), we apply a learnable linear projection:

$$Z = W_p H_G + b_p \quad (3)$$

where $W_p \in \mathbb{R}^{d_{T5} \times d_{graph}}$ and $b_p \in \mathbb{R}^{d_{T5}}$ are the weights and bias of the projection layer. The resulting tensor Z is fed directly into the T5 decoder as **encoder_hidden_states**. Effectively, the graph embeddings act as a "soft prompt," conditioning the generation mechanism via the cross-attention layers of the Transformer.

3.4.2 Position-Aware Weighted Loss

Standard language modeling minimizes the negative log-likelihood (NLL) uniformly across the sequence. However, in molecular captioning, the most critical information—the IUPAC name—is almost invariably located at the beginning of the sentence (e.g., "**2-chlorophenol** is a member of...").

To mitigate the "Naming Problem" where the model generates a correct description for the wrong molecule, we designed a custom Position-Aware Weighted Loss. Let $Y = (y_1, \dots, y_T)$ be the target caption sequence. The loss $\mathcal{L}$ is defined as a weighted average of the per-token cross-entropy:

$$\mathcal{L}(\theta) = - \frac{1}{\sum_{t=1}^T w_t} \sum_{t=1}^T w_t \cdot \log P(y_t | y_{<t}, Z; \theta) \quad (4)$$

Here, $P(y_t|\dots)$ is the probability distribution over the vocabulary predicted by T5, and w_t is a time-dependent weight factor defined as:

$$w_t = \begin{cases} \lambda_{\text{name}} & \text{if } t \leq 30 \\ 1.0 & \text{otherwise} \end{cases} \quad (5)$$

In our experiments, we set $\lambda_{\text{name}} = 5.0$. This modification forces the optimization process to assign five times more importance to the first 30 tokens, penalizing errors in the molecule’s name significantly more than syntax errors in the subsequent description.

3.4.3 Metric-Based Checkpointing

While the training loss serves as the optimization objective, it acts as a proxy that does not perfectly correlate with generation quality. To prevent overfitting to the training set’s specific phrasing, we implemented a metric-based checkpointing strategy. At the end of each epoch, we generate captions for the validation set using beam search and compute the BLEU score. The model parameters θ^* are updated only if the validation performance improves:

$$\theta^* \leftarrow \theta_t \iff \text{BLEU}_{\text{val}}(\theta_t) > \max_{k < t} \text{BLEU}_{\text{val}}(\theta_k) \quad (6)$$

This ensures that the final saved model is selected based on its generative capability rather than minimal perplexity.

4 Proposed Method II : Retrieval-based Graph-to-Text with Soft Editing

4.1 Contrastive Graph-Text Retrieval with Soft Multi-layer Pooling

We formulate molecular graph captioning as a cross-modal retrieval problem, where molecular graphs and textual descriptions are embedded into a shared latent space. The objective is to align each graph with its corresponding caption while pushing apart non-matching pairs. This formulation is particularly well suited to the limited-data regime of the task and directly optimizes for lexical similarity, which is strongly favored by BLEU-based evaluation.

Molecular graphs are encoded using a deep GINE architecture, utilizing the exact same Multi-Feature Embedding for atoms and bonds as detailed in Section 3.1. Rather than relying on a single graph-level representation extracted from the final layer, we adopt a multi-layer pooling strategy inspired by hierarchical representations. Specifically, after each GINE layer, we apply a global pooling operation to obtain an intermediate graph-level embedding. These embeddings capture structural information at different depths of the message-passing process, ranging from local chemical environments in early layers to more global molecular structure in deeper layers.

Formally, let $h_v^{(l)}$ denote the representation of node v after the l -th GINE layer, computed as described in Equa-

tion (2). We compute a pooled representation at each layer using a global add pooling operator:

$$g^{(l)} = \text{Pool} \left(\{h_v^{(l)} \mid v \in \mathcal{V}\} \right), \quad (7)$$

and concatenate these representations to form the final graph embedding:

$$g = \text{Concat} \left(g^{(1)}, \dots, g^{(L)} \right). \quad (8)$$

A linear projection is then applied to map this concatenated vector to the shared embedding dimension.

This *soft multi-layer pooling* strategy is used in place of explicit hierarchical pooling methods such as DiffPool. While DiffPool performs graph coarsening by learning soft cluster assignments, it inevitably compresses and aggregates node-level features. In our setting, atom- and bond-level features are extremely informative and must be preserved throughout the encoding process. Empirically, we found that explicit graph coarsening led to a loss of chemically relevant information and degraded retrieval performance. In contrast, multi-layer pooling preserves the full feature richness at each depth while still providing a hierarchical view of the molecular structure.

Textual descriptions are encoded using a pre-trained MiniLM sentence encoder. MiniLM is specifically designed to produce high-quality sentence embeddings and is widely used for semantic search, sentence similarity, clustering, and retrieval-augmented generation.

The text encoder is kept frozen during training. Unfreezing the encoder did not result in meaningful performance improvements, while significantly increasing training time. In practice, training with a frozen text encoder requires approximately 3 minutes per epoch, whereas unfreezing the encoder increases this cost to more than one hour per epoch. Given the absence of empirical gains, freezing the text encoder represents a favorable trade-off between performance and efficiency.

The resulting graph and text embeddings are L2-normalized and compared using cosine similarity, forming the basis of the contrastive training objective described in the next section.

4.2 Contrastive Training Objective

To align molecular graphs and textual descriptions in a shared embedding space, we adopt a contrastive learning framework. Given a batch of B paired samples, the objective is to maximize the similarity between matching graph-caption pairs while minimizing similarity to all other non-matching pairs within the batch.

Let $I_g \in \mathbb{R}^{B \times d}$ and $I_t \in \mathbb{R}^{B \times d}$ denote the L2-normalized graph and text embeddings, respectively. Similarity between graphs and texts is computed using cosine similarity, implemented as a dot product between normalized embeddings. Following CLIP-style models, we introduce a learnable temperature parameter s to control the sharpness of the similarity distribution. The similarity logits are defined as:

$$\text{logits} = \exp(s) \cdot I_g I_t^\top. \quad (9)$$

We optimize a symmetric contrastive cross-entropy loss:

$$\mathcal{L}_{CE} = \frac{1}{2} \left(\mathcal{L}_{CE}(\text{logits}, y) + \mathcal{L}_{CE}(\text{logits}^\top, y) \right), \quad (10)$$

where y denotes the identity labels enforcing correct alignment between graphs and captions within each batch. This symmetric formulation ensures that both graph-to-text and text-to-graph retrieval are optimized jointly.

We also experimented with Circle Loss, a metric learning objective that explicitly models the relative importance of positive and negative similarities. While Circle Loss led to slightly faster convergence during early training, it did not provide better retrieval performance compared to contrastive cross-entropy. Given that the graph-text retrieval model converges rapidly in practice (under 15 minutes in total), the marginal difference in convergence speed was not sufficient to justify replacing the simpler and more stable contrastive cross-entropy loss. As a result, contrastive cross-entropy was retained as the training objective in the final system.

4.3 Inference and Reranking Strategies

At inference time, all training molecular graphs are embedded once and stored in a vector index. Given a query graph embedding g , we retrieve the top- K nearest neighbors using cosine similarity (dot product between L2-normalized embeddings). This *graph-only similarity* is the backbone retrieval mechanism used to construct an initial pool of candidate captions. We select $K = 10$, tuned empirically on the validation set.

Let $\{c_i\}_{i=1}^K$ denote the retrieved candidate captions and $\{s_i^G\}_{i=1}^K$ their graph similarity scores (retrieval scores). Several candidates are often structurally close yet differ in subtle textual details. We therefore evaluated multiple reranking strategies to select the final caption.

Graph-Text Similarity Reranking: A first approach reranks candidates using direct graph-text similarity in the shared embedding space. Let t_i be the embedding of caption c_i (MiniLM), and g the query graph embedding. We score candidates as:

$$s_i^{GT} = \langle t_i, g \rangle, \quad (11)$$

and select $\arg \max_i s_i^{GT}$.

Lexical Centroid Rerankin: We also tested a purely text-based reranker that selects the caption most similar to the centroid of candidate text embeddings. This favors captions that are linguistically representative of the retrieved set, but it ignores direct alignment with the query graph.

Hybrid Reranking: Our strongest reranking family combines graph-based evidence with text-based evidence. Concretely, we define a hybrid score:

$$s_i = \alpha s_i^G + (1 - \alpha) s_i^{lex}, \quad (12)$$

where s_i^{lex} is a lexical representativeness score (similarity of t_i to the candidate centroid), and α controls the trade-off. This strategy consistently outperforms purely graph-based or purely text-based reranking.

Minimum Bayes Risk (MBR): We experimented with MBR-style selection, where the best caption minimizes expected risk with respect to other candidates. In our implementation, the pairwise similarity between captions is estimated using CHRF as a sequence-level proxy for BLEU-like overlap. Both a plain MBR variant and a graph-weighted variant were evaluated. In practice, MBR did not outperform the hybrid reranking.

Distance-based Candidate Pruning: In addition to reranking, we apply a pruning step to discard candidates that are too far from the best retrieved neighbor according to graph similarity. Let $s_{\max}^G = \max_i s_i^G$. We keep only candidates satisfying:

$$s_i^G \geq s_{\max}^G - \delta, \quad (13)$$

with $\delta = 0.02$ (selected empirically on the validation set), and rerank only the remaining candidates using the hybrid score. This pruning step improves robustness by removing weakly related captions that may otherwise be selected due to lexical effects.

Table 1: Impact of different reranking strategies on validation performance. Reported values are indicative and reflect relative trends rather than exact scores.

Method	BLEU-4	BERTScore
Top-1 Retrieval (no reranking)	≈ 0.37	≈ 0.91
Hybrid Reranking ($K = 10$)	≈ 0.385	≈ 0.92
Hybrid + Pruning (final)	$\approx \mathbf{0.394}$	$\approx \mathbf{0.92}$
MBR Reranking	≈ 0.37	≈ 0.91

Quantitative Impact: From a quantitative perspective, the different reranking strategies mainly affect BLEU score, while BERTScore remains consistently high across all variants. Using the top-1 retrieved caption without reranking yields a BLEU score of approximately 0.37 with a BERTScore around 0.91. Introducing hybrid reranking with $K = 10$ improves BLEU to about 0.385, while slightly increasing BERTScore.

The best performance is obtained with hybrid reranking combined with distance-based pruning, reaching a BLEU score close to 0.394 and a BERTScore around 0.92. In contrast, MBR-based reranking does not improve over the baseline retrieval and yields BLEU scores comparable to the no-reranking setting, despite maintaining a high BERTScore.

These results indicate that reranking primarily refines lexical alignment, which is directly reflected by BLEU, whereas semantic similarity, as measured by BERTScore, remains high for most retrieval-based methods.

4.4 Graph-guided Soft Editing

Beyond retrieval and reranking, we apply a lightweight *soft editing* step to further improve caption accuracy. The motivation is that retrieved captions are often globally correct but may contain a small number of local inconsistencies with respect to the target molecular graph. In practice, these inconsistencies frequently concern charge-related descriptors and chemical species categories, which

are directly encoded in the graph but imperfectly reflected in the retrieved text.

Our editing strategy is deliberately conservative and rule-based. It operates directly on the selected caption and leverages atom-level features from the query graph, in particular the formal charges. Based on these features, we enforce consistency for key terms such as *anion*, *cation*, *zwitterion*, and *conjugate acid/base*. For instance, captions describing a molecule as a *conjugate acid* are automatically corrected to *conjugate base* when the net molecular charge is negative, and vice versa. Similarly, the species designation is adjusted depending on the presence of positive and/or negative atomic charges.

This editing step does not alter the global sentence structure and avoids any generative rewriting. Its objective is solely to correct high-impact lexical errors that are heavily penalized by BLEU. Empirically, adding soft editing on top of the hybrid reranking with pruning yields a consistent gain of approximately **+0.01 BLEU** on the validation set, while leaving BERTScore essentially unchanged. This confirms that soft editing mainly improves lexical precision rather than semantic content, and complements retrieval-based methods in a cost-effective and robust manner.

4.5 Cross-Encoder for Graph-Text Matching

Finally, we investigated a Cross-Encoder model to explicitly capture fine-grained interactions between graph and text modalities. Operating on top of the retrieval stage, this model scores graph-caption pairs by processing their concatenated embeddings and element-wise differences through an MLP. It was trained using a margin ranking loss while keeping the underlying encoders frozen. Although the model successfully learned to distinguish positive pairs, it failed to outperform the simpler hybrid reranking strategy in terms of BLEU score. This suggests that once a strong retrieval backbone is established, remaining errors are primarily lexical rather than due to a lack of expressive modeling power. Consequently, we excluded this computationally heavier component from the final pipeline.

5 Results & Conclusion

In this section, we compare the performance of the different approaches developed during the challenge. The primary comparison metric is the score obtained on the public Kaggle leaderboard, which aggregates BLEU and BERTScore evaluations on the held-out test set.

Table 2: Comparison of methods on the Kaggle Leaderboard.

Method	Kaggle Score $\uparrow$
Rule-based Editing	≈ 0.576
Simple CLIP+RAG+LLM	≈ 0.583
"From-scratch" Transformer	≈ 0.601
GINE + T5 (Method I)	≈ 0.618
RAG + Soft Editing (Method II)	≈ 0.619

Analysis of Baselines and Exploratory Methods:

The *Simple CLIP+RAG+LLM* approach yielded relatively low performance (≈ 0.583). Despite utilizing GPT-2, the naive retrieval strategy led the model to largely ignore the graph embedding, often regurgitating retrieved captions without adapting them to the specificities of the new molecule. Similarly, the *"From-scratch" Transformer* (≈ 0.601) initially showed promise but ultimately plateaued. As discussed, its lack of pre-trained linguistic knowledge resulted in grammatical errors that penalized the score. The *Rule-based Editing* approach (≈ 0.576) performed poorly on its own, as its rigid templates failed to generalize to unseen chemical structures and situations not observed during training.

Performance of Proposed Methods: The two main architectures detailed in this report achieved the highest scores.

- **GINE + T5 (≈ 0.618):** This generative approach successfully bridged the gap between structural encoding and text generation. By leveraging the pre-trained T5 decoder, it eliminated the grammatical issues of the "from-scratch" model. Furthermore, the custom position-aware loss proved critical in correctly generating IUPAC names, a major factor in the improved score.
- **RAG + Soft Editing (≈ 0.619):** This method achieved the best overall performance. By framing the task as a retrieval problem, we maximized lexical overlap with ground-truth captions (high BLEU). The "Soft Editing" step was the key differentiator, refining the retrieved captions by correcting specific chemical properties (e.g., charge, ion status) using rule-based logic derived from the graph. This hybrid strategy combined the fluency of human-written captions with the precision of graph-derived attributes, proving to be the most robust solution for this challenge.

Conclusion: In conclusion, while fully generative models offer greater flexibility, the constrained nature of the dataset favored retrieval-based strategies enhanced by domain-specific logic. Our experiments demonstrate that combining a strong structural encoder (GINE) with either a pre-trained language model (T5) or a sophisticated retrieval mechanism yields the most effective results for Molecular Graph Captioning.

References

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Appendix

Table 3: Statistical Analysis of Tokenized Caption Lengths (T5 Tokenizer). Setting $L_{max} = 274$ covers 99% of the validation set.

Metric	Training Set	Validation Set
Count	31,008	1,000
Mean	108.63	111.13
Median (p50)	100.00	101.00
Min	29	35
Max	920	444
<i>Percentile 90%</i>	168.00	173.10
<i>Percentile 95%</i>	192.00	202.05
Percentile 99%	250.00	274.14

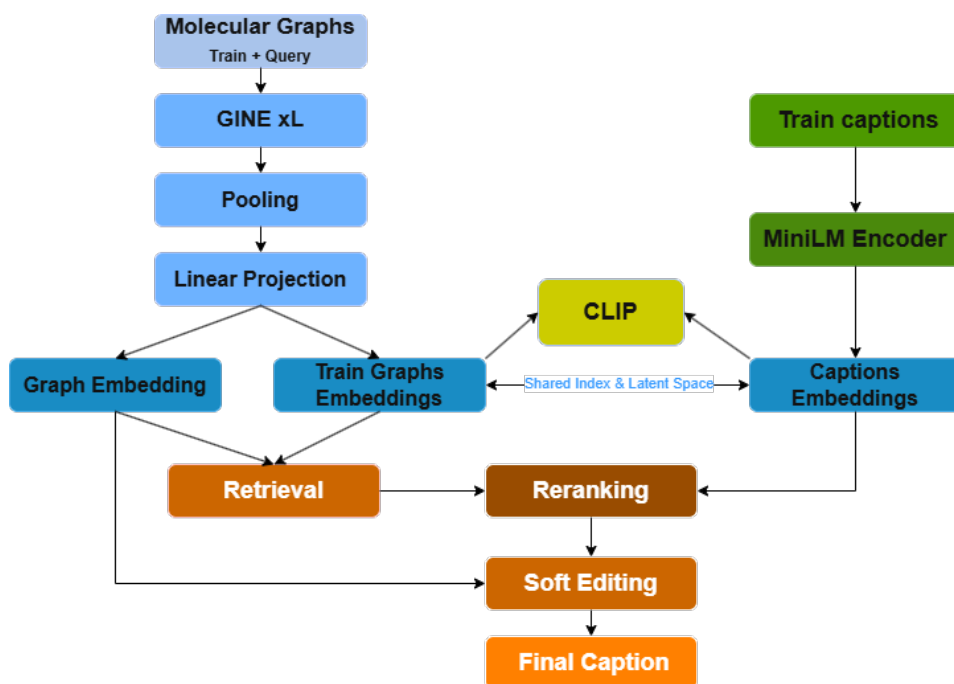


Figure 1: Retrieval Model Architecture